

Skills Summary

Multistep Spatial and Coordinate Problems

Use this reference while completing your worksheet or when studying.

Skill 1 — A missing side of a right triangle

The theorem: $a^2 + b^2 = c^2$, with c the hypotenuse (the longest side, opposite the right angle).

Missing the hypotenuse: add, $c = \sqrt{a^2 + b^2}$ — e.g. legs 3 and 4 give $\sqrt{25} = 5$.

Missing a leg: subtract, $a = \sqrt{c^2 - b^2}$. Which one you need is decided by whether the longest side is already known.

Example: Hypotenuse 26, one leg 10. Find the other leg.

Step 1. The longest side is given, so the missing side is a leg — subtract the squares.

Step 2. $\sqrt{26^2 - 10^2} = \sqrt{676 - 100} = \sqrt{576} \Rightarrow 24$

Try it: Hypotenuse 37, one leg 12. Find the other leg.

check: 35

Skill 2 — Distance between two points on a grid

Rule: the horizontal gap and the vertical gap are the two legs, and the segment joining the points is the hypotenuse. So a distance is always found by *adding*:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Squaring removes any minus sign, so it does not matter which point you subtract first.

Example: Find the distance from (2, 1) to (10, 7).

Step 1. Two points and a length wanted, so build the right triangle from the two gaps and add.

Step 2. Gaps: $10 - 2 = 8$ and $7 - 1 = 6$, so $d = \sqrt{64 + 36} = \sqrt{100} \Rightarrow d = 10$

Try it: Find the distance from (1, -2) to (13, 3).

check: 13

Skill 3 — The diagonal of a box, in two stages

Stage 1: the diagonal across the base, $d = \sqrt{\ell^2 + w^2}$.

Stage 2: treat that answer as a leg and the height as the other leg, $D = \sqrt{d^2 + h^2}$.

Doing both stages at once gives $D = \sqrt{\ell^2 + w^2 + h^2}$ — the same thing, with the first square root never written down.

Example: A box has a base 6 by 8 and a height of 24. Find both diagonals.

Step 1. A box needs two stages: flat across the base first, then upwards using that as a leg.

Step 2. $d = \sqrt{6^2 + 8^2} = \sqrt{100}$, then $D = \sqrt{10^2 + 24^2} = \sqrt{676}$. $\Rightarrow d = 10$ and $D = 26$

Try it: A box has a base 4 by 3 and a height of 12. Find the diagonal of the box.

check: 13

Skill 4 — Multistep journeys and outlines

Method: find the missing side first, then do the ordinary arithmetic the question really asks for — a total, a difference, a perimeter.

Going along two sides of a rectangle costs $\ell + w$; cutting the diagonal costs $\sqrt{\ell^2 + w^2}$, which is always less. The saving is the difference of the two.

Example: A field is 9 m by 12 m. How much shorter is the diagonal than walking two sides?

Step 1. The question is a difference, so find the diagonal first and then subtract it from the two-side route.

Step 2. Diagonal = $\sqrt{9^2 + 12^2} = \sqrt{225} = 15$; two sides cost $9 + 12 = 21$; the saving is $21 - 15$. $\Rightarrow$ diagonal = **15** m, saving = **6** m

Try it: A field is 5 m by 12 m. How many metres does the diagonal save?

check: 4

(!) **Watch out:** an answer that comes out *longer* than the hypotenuse you started with means the squares were added when they should have been subtracted.